K-INJECTIVITY DESCENDS ALONG AN EPIMORPHISM OF RINGS

THE STACKS PROJECT AUTHORS

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Editorial note. The author's proof is terse but complete: it names the exact identification or cited result that settles the statement, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- Homotopy classes of maps between complexes of modules over a quotient coincide over the ring and over the quotient when the ring map is an epimorphism. Algebra, Lemma \ref{algebra-lemma-epimorphism-modules}, cited at proof lines 13949-13952. Supplies the identification that transfers the K-injectivity property.

Source reference. Original lemma 0916: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11434–11434. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 08YS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11350–11350. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to S$ be an epimorphism of rings. Let $I^\bullet$ be a complex of S-modules. If $I^\bullet$ is K-injective as a complex of R-modules, then $I^\bullet$ is a K-injective complex of S-modules.*

Proof. This is true because $\mathrm{Hom}_{K(R)}(N^\bullet, I^\bullet) = \mathrm{Hom}_{K(S)}(N^\bullet, I^\bullet)$ for any complex of S-modules $N^\bullet$, see Algebra, Lemma 08YS. □

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